

Median

Definition

The **Median** is a **measure of central tendency**. It is the single value that represents the **exact middle** of a dataset that has been arranged in ascending (smallest to largest) or descending (largest to smallest) order.

The median is also known as the **50th percentile** because it divides the dataset into two equal halves:

- 50% of the data values are **less than or equal to** the median.
- 50% of the data values are **greater than or equal to** the median.

The median is often preferred over the mean (average) when a dataset has **extreme values (outliers)**, because the median is not affected by them.



Formula For Both Odd And Even

The "formula" for the median is first about finding its **position** in an ordered dataset. Let N be the total number of observations (the count of all data points).

Step 1: Order the data from smallest to largest.

Step 2: Find the position of the median using the formula:

$$\text{Median Position} = \frac{N + 1}{2}$$

Case 1: N is Odd (e.g., $N = 5, 7, 9$)

- When N is odd, the formula $\frac{N+1}{2}$ gives a whole number.

- **Formula:** The median is the value at this *exact position*.
- **Example:** If $N = 7$, the position is $\frac{7+1}{2} = 4$. The median is the **4th value** in the ordered list.

Case 2: N is Even (e.g., $N = 6, 8, 10$)

- When N is even, the formula $\frac{N+1}{2}$ gives a decimal (e.g., 3.5, 5.5).
- A position like "3.5" means the median lies *between* the 3rd and 4th values.
- **Formula:** The median is the **average of the two middle values** (the values on either side of the decimal position).
- **Example:** If $N = 6$, the position is $\frac{6+1}{2} = 3.5$. The median is the **average of the 3rd and 4th values**.



Examples

Ungrouped Data

This is a raw list of numbers.

Example 1: N is Odd

- **Data:** 19, 12, 16, 25, 14
- **Step 1 (Order):** 12, 14, 16, 19, 25
- **Step 2 (N):** $N = 5$ (Odd)
- **Step 3 (Position):** $\frac{5+1}{2} = \frac{6}{2} = 3$ rd position.
- **Step 4 (Median):** The 3rd value in the ordered list is **16**.

Example 2: N is Even

- **Data:** 30, 10, 20, 50, 40, 60
- **Step 1 (Order):** 10, 20, 30, 40, 50, 60
- **Step 2 (N):** $N = 6$ (Even)
- **Step 3 (Position):** $\frac{6+1}{2} = \frac{7}{2} = 3.5$ th position.

- **Step 4 (Median):** The position 3.5 is between the 3rd and 4th values.
 - 3rd value = 30
 - 4th value = 40
 - Median = $\frac{30+40}{2} = \frac{70}{2} = 35$



Discrete Frequency Distribution

This is when data is presented in a table showing each value (x) and how many times it occurs (frequency, f).

Procedure:

1. Calculate the **Cumulative Frequency (cf)**. This is a running total of the frequencies.
2. Find the total frequency, $N = \sum f$.
3. Find the median position using $\frac{N+1}{2}$.
4. Locate the cf value that is **just greater than or equal to** the median position.
5. The median is the **x value** (the data value, not the frequency) corresponding to that cf .

Example:

x (Marks)	f (Students)	cf (Cumulative Freq.)
10	2	2
20	5	$2 + 5 = 7$
30	8	$7 + 8 = 15$
40	3	$15 + 3 = 18$
50	2	$18 + 2 = 20$

- **Step 2 (N):** $N = \sum f = 20$ (Even)
- **Step 3 (Position):** $\frac{20+1}{2} = \frac{21}{2} = 10.5$ th position.
- **Step 4 (Locate cf):** We need to find the cf just greater than or equal to 10.5.
 - cf of 2 is too small.

- cf of 7 is too small.
- cf of 15 is the first one greater than 10.5.
- **Step 5 (Median):** The x value corresponding to the cf of 15 is 30.
 - (This means the 8th, 9th, 10th, 11th, 12th, 13th, 14th, and 15th students all scored 30. Since we are looking for the 10.5th value, it falls in this group.)



Continuous Frequency Distribution

This is when data is grouped into class intervals (e.g., 10 – 20, 20 – 30). We first find the *Median Class* and then use a formula to estimate the median value within that class.

Procedure:

1. Calculate the **Cumulative Frequency (cf)**.
2. Find the total frequency, $N = \sum f$.
3. Find the median *class location* using $\frac{N}{2}$. (Note: We use $\frac{N}{2}$ here, not $\frac{N+1}{2}$, as per standard convention for continuous data).
4. Find the **Median Class**: This is the class interval whose cf is **just greater than or equal to** the $\frac{N}{2}$ value.
5. Apply the Median Formula:

$$\text{Median} = L + \frac{\left(\frac{N}{2} - cf\right)}{f} \times i$$

Where:

- L = **Lower limit** of the Median Class.
- N = **Total frequency** ($\sum f$).
- cf = **Cumulative frequency of the class preceding (before) the Median Class**.
- f = **Frequency** of the **Median Class** itself.
- i = **Class width** (Upper limit - Lower limit) of the Median Class.

Example:

Class Interval (Marks)	f (Students)	cf (Cumulative Freq.)
0 – 10	5	5
10 – 20	10	$5 + 10 = 15$
20 – 30	20	$15 + 20 = 35$
30 – 40	8	$35 + 8 = 43$
40 – 50	7	$43 + 7 = 50$

- **Step 2 (N):** $N = 50$
- **Step 3 (Location):** $\frac{N}{2} = \frac{50}{2} = 25$ th position.
- **Step 4 (Median Class):** We look for the cf just greater than or equal to 25.
 - cf of 5 is too small.
 - cf of 15 is too small.
 - cf of **35** is the first one greater than 25.
 - Therefore, the **Median Class** is **20 – 30**.
- **Step 5 (Apply Formula):**
 - $L = 20$ (Lower limit of 20 – 30)
 - $N = 50$
 - $cf = 15$ (The cf of the class *before* 20 – 30)
 - $f = 20$ (The frequency of the 20 – 30 class)
 - $i = 10$ (30 – 20)

$$\text{Median} = 20 + \frac{\left(\frac{50}{2} - 15\right)}{20} \times 10$$

$$\text{Median} = 20 + \frac{(25 - 15)}{20} \times 10$$

$$\text{Median} = 20 + \frac{10}{20} \times 10$$

$$\text{Median} = 20 + 0.5 \times 10$$

$$\text{Median} = 20 + 5$$

$$\text{Median} = \mathbf{25}$$

The median mark is 25. This value makes sense as it falls within our 20 – 30 median class.